INCH-POUND

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12 JULY 1997
SUPERSEDING
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12 January 1994

PERFORMANCE SPECIFICATION

BOOTH, CLEANING, WATER BASE SOLVENT SPRAY-ELECTRONIC MEASURING INSTRUMENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of floor mounted, manually operated, nonrecirculating water base solvent spray booth for the cleaning of electronic measuring instruments.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Road, Kelly AFB, TX 78241-6424 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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2.2.1 Government documents

2.2.2 Specifications, standards and handbooks. The following specifications, standards and handbooks form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-C-87937 Cleaning Compound, Aerospace Equipment

(Unless otherwise indicated, copies of Federal and Military Specifications, Standards, and Handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified,the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WELDING SOCIETY, INC.

AWS D1.1 Structural Welding Code Steel

AWS D1.2 Structural Welding Code Aluminum

(Application for copies of AWS publications should be addressed to the American Welding Society, Inc., 550 N. W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 National Electrical Code

NFPA 79 Electrical Standard for Industrial Machinery

NFPA 91 Standards for Exhaust Systems for Air Conveying of Materials

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

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2.4 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. This specification provides for first article testing. (see paragraph 4.3 and 4.4)

3.2 Function. This booth shall clean electronic instruments, as normally used in Air Force Precision Measurement Equipment Laboratories, by spraying the item with a water based cleaning solution, followed by a clear water rinse, and the removal of moisture with an air blast. The booth shall accommodate instruments up to 24 inches by 24 inches by 24 inches in size and weighing no more than 150 pounds.

3.3 Description. The booth shall incorporate features that allow the operator to clean an item through a full 360 degree rotation manually without the operator physically repositioning it. The operator shall stand outside the booth to operate the cleaning, rinsing, and drying nozzles as well as rotating the item being cleaned. All vapors shall be externally exhausted and liquids drained. Lighting shall be provided so that the work piece is shadowless. Cleaning and rinsing nozzles shall be adjustable so that the fineness or coarseness of the spray is controllable. A reservoir shall be included to contain sufficient cleaning solution for four hours of continuous operation. All controls and indicators shall be within reach and view from the operators work location and nozzles shall be self storing. Mix ratio of the cleaning solution to water shall be adjustable from at least 1 part solution to 3-6 parts water. Operating air pressure shall be adjustable within the 20 to 80 psig range minimum. Hoses serving spray and air blast guns shall withstand a burst test pressure of not less than four times the maximum pressure developed by a stall or bypass pressure.

3.4 Interfaces.

a. Electrical power; 120 Volt AC, 60 Hz, 1 Phase via junction box.

b. Tap water; 60 psig via three-fourths (3/4) inch IPS nipple.

c. Shop compressed air; 200 psig via three-eighths (3/8) inch IPS nipple.

d. Waste drain; via three-fourths (3/4) inch IPS nipple.

e. Exhaust air venting; via duct flange meeting the requirements of NFPA Standard No 91.

f. Compatible with MIL-C-87937, type I, cleaning compound.

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3.5 General. The booth shall be furnished complete so that the booth may used and operated as intended. It shall be free of defects , such as air or liquid leaks. Electrical wiring and components shall adhere to National Fire Protection Association Standard No. 70 and be protected where ever possible. Workmanship and construction shall be of good commercial quality and where applicable welding shall be in accordance with AWS D1.1 and AWS D1.2.

3.6 Weight. Dry weight of the booth shall not exceed 300 pounds.

3.7 Stand or base. The stand or base shall allow for anchoring to the floor and position the work surface at a convenient height for the operator and shall provide space for the storage of cleaning concentrate containers.

3.8 Assembly. The booth shall not be disassembled to the extent that it would require special skills or tools to reassemble.

3.9 Instruction plate. An instruction plate shall be affixed to the booth that contains operating instructions and warnings, if any, within easy view from the operators work location.

3.10 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Environmentally preferable products are those products or services that have a lesser negative effect on human health or the environment when compared with competing products or services that serve the same purpose. This comparison should use principles recommended in guidance by the EPA, and may consider raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, or disposal of the product or service.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. The government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification . The inspection and testing of the booth shall be classified as follows:

a. First article inspection (see paragraph 4.3).

b. Quality Conformance inspection (see paragraph 4.4).

4.3 First article. The first article test shall consist of all tests described under test methods.

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4.3.1 First article test sample. When a first article inspection is required, the item to be tested shall be as specified in paragraph 4.5 Test method. The sample shall be identical in all respects to the booths to be produced as production items.

4.3.2 First article test report. Upon completion of the first article inspection and tests, a test report shall be prepared in accordance with contractor format. The report shall include the results of the tests conducted under examination (4.5.1) and operation (4.5.2) as well as checklists used during the examination and operation.

4.4 Quality conformance inspection. Inspection of each production booth shall consist of examination (4.5.1) and operation (4.5.2).

4.5 Test methods.

4.5.1 Examination. The booth, including its components, shall be inspected to determine compliance with the requirements herein with respect to design and configuration. All weldments for compliance to AWS D1.1 and AWS D1.2. The structure and panels shall be free from kinks and waviness, all edges shall be smooth, and all hardware shall be properly secured and holes properly aligned. Wiring and electrical components for security, protection from the operating environment shall conform to NFPA Standard No. 70 and 79. First article examination, if required, shall be accomplished by means of a contractor developed checklist which shall list booth characteristics subject to examination, method of verification and the results obtained. Noncompliance shall be cause for rejection.

4.5.2 Operation. The booth shall be hooked up to the appropriate interfaces and checked for air and liquid leakage at 50% above maximum rated pressures. All nozzles shall deliver at appropriate pressures and flow rates, and when adjustable the adjustment shall operate easily and hold its setting until manually changed. A sample item weighing at least 150 pounds shall be used to test the rotating apparatus for safety of unbalanced positioning, and that the operator can rotate the item being cleaned without interrupting the cleaning, rinsing or drying operation. The exhaust system shall vent all vapors in an efficient manner. Drainage shall be complete, without any pooling inside the booth. Solvent mixing ratios shall be verified by a time and usage test.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory) .

6.1 Intended use. The booth is intended for the cleaning of precision electronic measuring instruments by the removal of conductive contamination by means of a water base cleaning solution spray, water rinse spray and air blast for water removal. Final drying of the instrument so cleaned is to be effected by the use of an Oven, Infrared Drying (low temperature). (NSN 4430-00-472-3342)

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Preservation, packaging and packing required.

6.3 Subject term (key word) listing.

Enclosure
Air dry gun
Regulator
Spray Gun
Exhausted
Turntable

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force-99
Army - AL

Preparing Activity:
Air Force-82

Agent Activity
Air Force -99

(Project 4940-0679)

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I RECOMMEND A CHANGE:	1. DOCUMENT	2. DOCUMENT DATE
	MIL-PFR-83831A	97/07/12

3. DOCUMENT

BOOTH, CLEANING, WATER BASE SOLVENT SPRAY-ELECTRONIC MEASURING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as*

5. REASON FOR

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
SA-ALC/TILDD	(1) Commercial (210) 925-6314	(2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Rd. Kelly AFB, TX 78241-	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041- Telephone (703) 756-2340 AUTOVON 289-2340	

DD FORM 1426, OCT 89*(EF-V1)* *(PerFORM PRO)* PREVIOUS EDITIONS ARE OBSOLETE.

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